Imam Abdulrahman Bin Faisal University
Deanship of Preparatory Year and Supporting Studies
Computer Department



Basics of Programming through Python

Getting started with Python

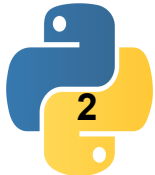
Introduction to Programming

COMP106

Term 1 | 2026-2027

Learning outcomes

- Describe the features of Python programming language.
- List Python IDEs.
- Differentiate between Python script mode and interactive mode.
- Explain how to install Python and Jupyter Notebook.
- Use Jupyter Notebook as the programming environment.
- Write a first Python program.



Features of Python (1)

- Python language is founded in the year 1991 by the developer Guido Van Rossum.
- Its most popular implementation is called CPython and is written in C programming language.
- Python is a high level language.
- It is a free and open source software that works on Linux, Mac, Windows, and various other platforms.

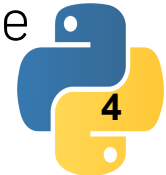
  ➔Python is portable and platform independent, means it can run on various operating systems and hardware platforms.



Features of Python (2)

- It is an interpreted language, as Python programs are executed by an interpreter.
- Python programs are easy to understand as they have a clearly defined syntax and relatively simple structure.
- Python is case-sensitive.

  ➔For example, NUMBER and number are not same in Python.



Features of Python (3)

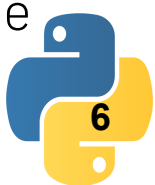
- Python has a rich library of predefined functions.
- Python is also helpful in Web development.

  ➔ Many popular Web services and applications are built using Python.
- Python uses indentation for blocks and nested blocks.



Python Interpreter

- The Python interpreter is a program that can read Python programming statements and execute them.
- When you install the Python language on your computer, one of the items that is installed is the Python interpreter.
- The Python interpreter can run Python programs that are saved in files or interactively execute Python statements that are typed at the keyboard.
- Python comes with a program named IDLE that simplifies the process of writing, executing, and testing programs.



Python IDE / IDLE

- Python IDLE (stands for Integrated Development and Learning Environment): an integrated development environment for Python, which has been bundled with the default implementation of the language.
- What is an IDE (stands for Integrated Development Environment)?

  ➔An IDE is a software application that provides comprehensive facilities to computer programmers for software development.
- Most modern IDEs have intelligent code completion.



Popular Python IDEs





























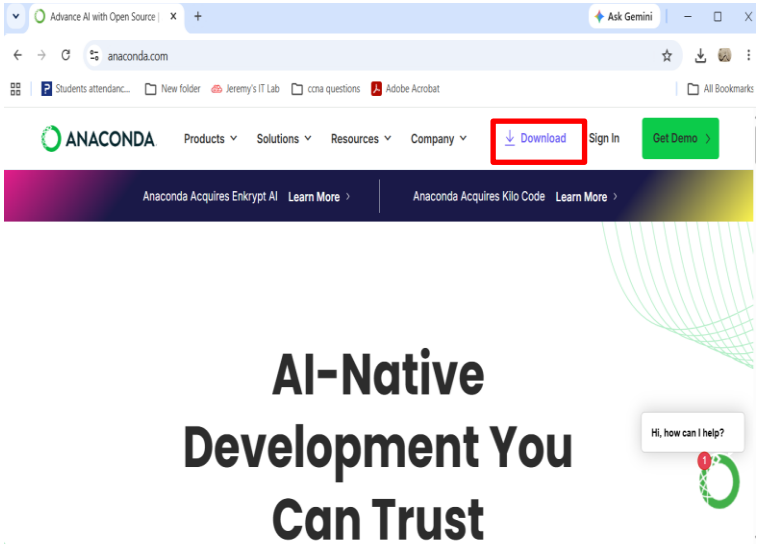


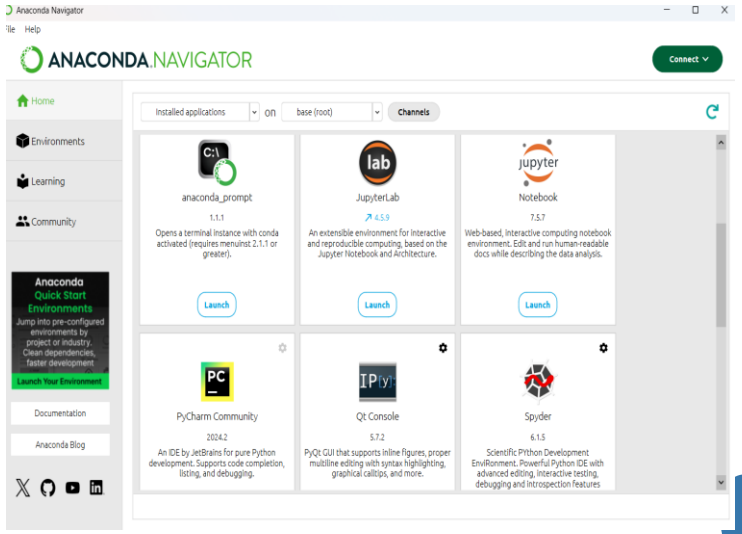


Installing Python and Jupyter Notebook IDE (1)

Installing Jupyter Notebook on Windows

- Go to the official Anaconda website and open the download page.
  https://www.anaconda.com/

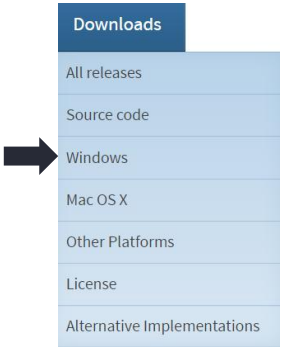


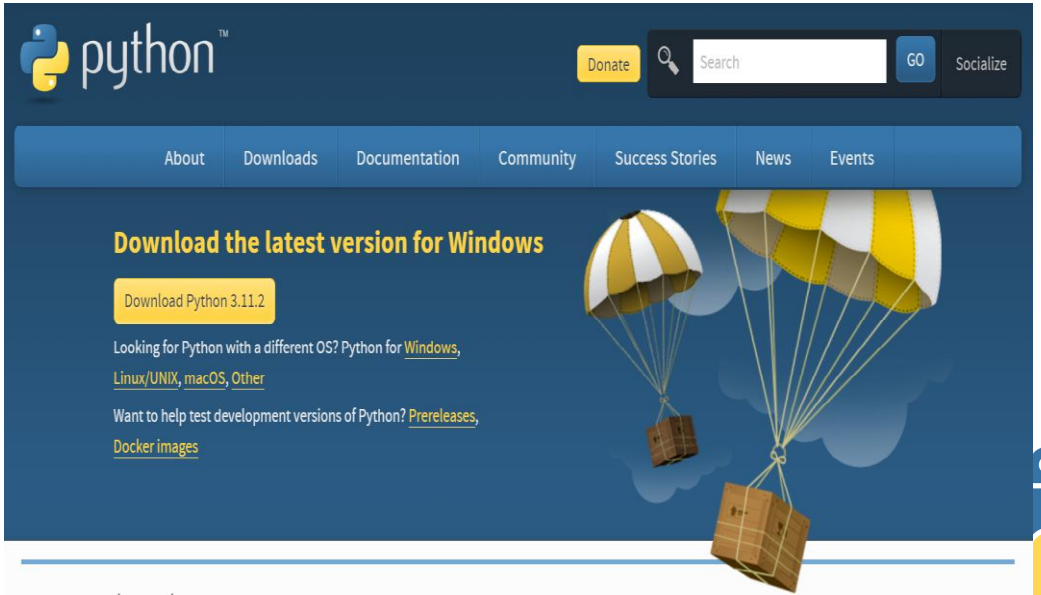


Installing Python and Jupyter Notebook IDE (1)

Installing Python on Windows

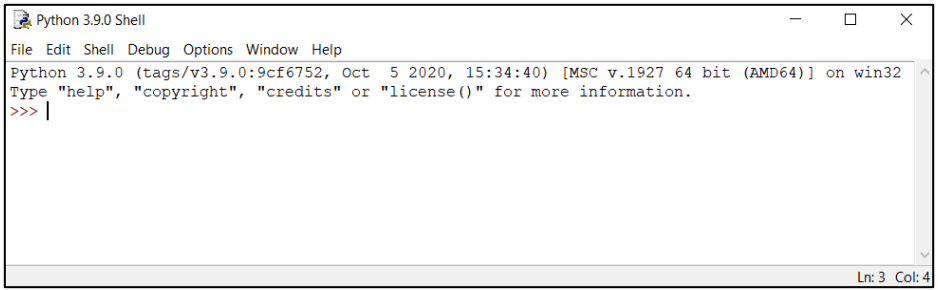
- Download the latest Python 3 version (64-bit graphical installer for most modern systems), from the official website: https://www.python.org/

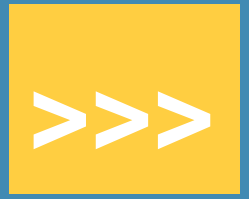




Running Python
Working with default Python distribution

- Now the Python interpreter is installed on your computer to run and execute python programs. You can use also any **online Python interpreter** (i.e. https://www.online-python.com/).
- The interpreter is also called Python shell.
- A sample screen of Python interpreter (or shell) is shown in the following figure.





- The symbol >>> is the Python prompt, which indicates that the interpreter is ready to take instructions.
- We can type commands or statements on this prompt to execute them using a Python interpreter.





Execution modes

- There are **two ways** to use the Python interpreter:
    1) Interactive mode: allows execution of individual statement instantaneously
    2) Script mode: allows us to write more than one instruction in a file called Python source code file that can be executed.



Interactive mode (1)

- To work in the interactive mode, we can simply type a Python statement on the **>>> prompt directly**.
- As soon as we press enter, the interpreter executes the statement and displays the result(s), as shown in the figure.

```
>>> 1+2
3
>>> 4-2
2
>>> "Hello" + "World"
'HelloWorld'
>>>
```

Interactive mode (2)

- Working in the interactive mode is convenient for testing a single line code for instant execution.
- But in the interactive mode, we cannot save the statements for future use and we have to retype the statements to run them again.



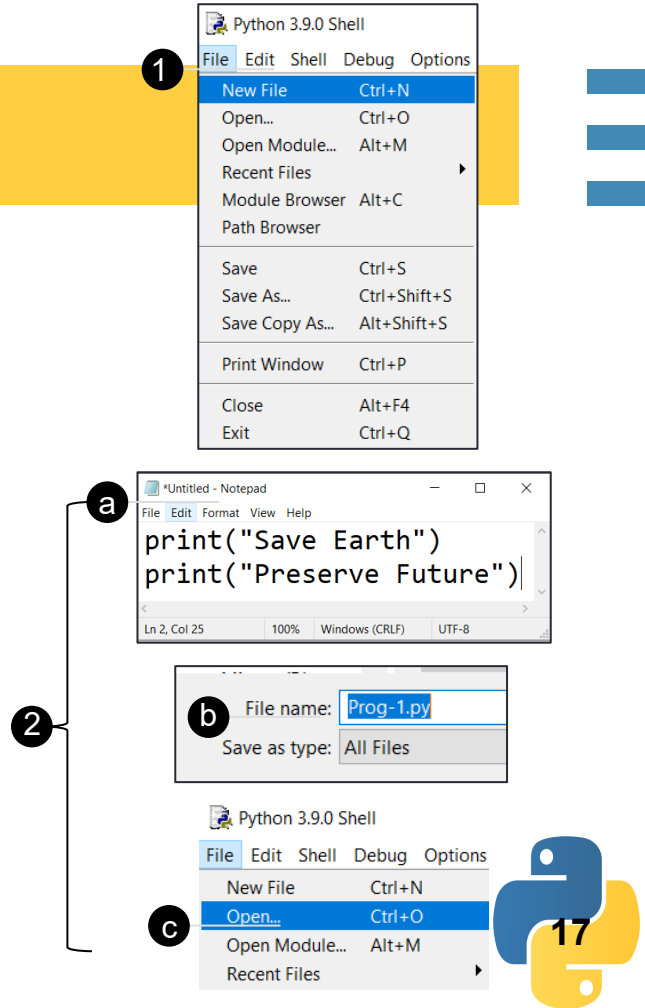
Script mode (1)

- In the script mode, we can write a Python program in a file, save it and then use the interpreter to execute it.
- Python scripts are saved as files where file name has extension ".py".
- By default, the Python scripts are saved in the Python installation folder.



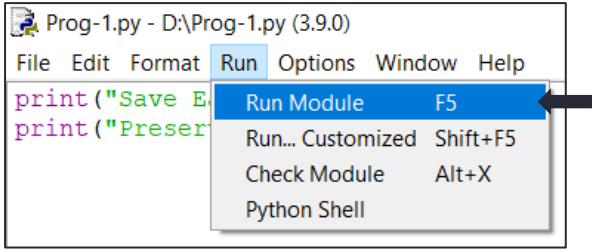
Script mode (2)

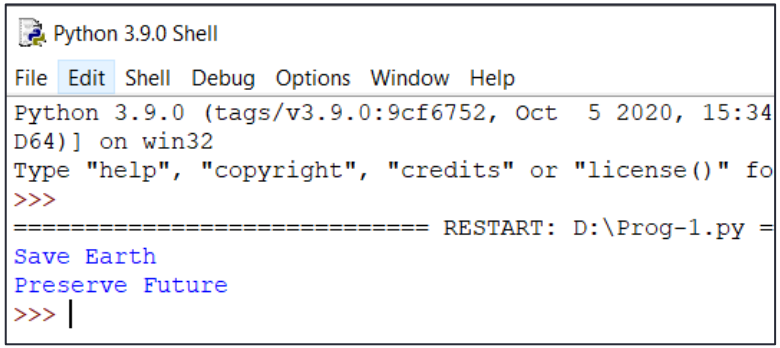
- To create a script file, we can either:
  1) Create a python file (.py) from IDLE ("**File**" ➔"**New File**"), and save it.
  2) **a)** Create your Python file (.py) using editor like **Notepad++**,
  b) save it an **.py** file, then
  c) open it directly from IDLE ("**File**" ➔"**Open**") as shown in the figure.



Script mode (3)

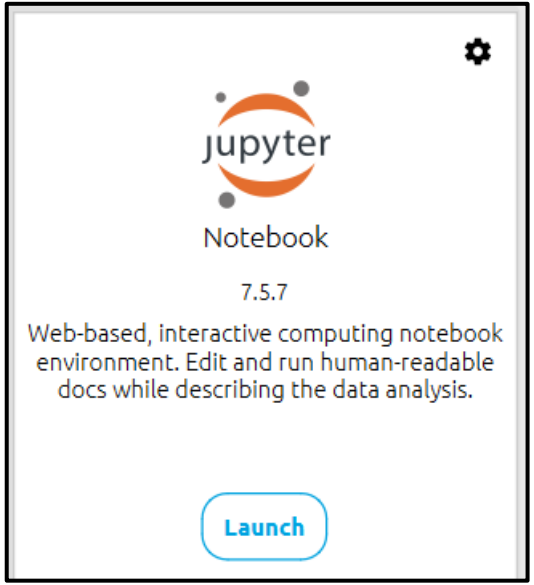
3) While working in the script mode, after opening the file, click on "**Run**"➔"**Run** *Module*" from the menu as shown in the figure.





Launching Jupyter NotebookJupyter Notebook (1)

- Open Anaconda Navigator.
  Find Jupyter Notebook in the list of applications.
  Click Launch to open Jupyter Notebook in your web browser.



To download it visit anaconda navigator: https://www.anaconda.com/

Python Keywords

- Keywords are **reserved words**.
- Each keyword has a specific meaning to the Python interpreter, and we can use a keyword in our program only for the purpose for which it has been defined.
- As Python is **case sensitive**, keywords must be written exactly as given in the table.

False	**class**	**finally**	**is**	**return**
None	**continue**	**for**	**lambda**	**try**
True	**def**	**from**	**nonlocal**	**while**
and	**del**	**global**	**not**	**with**
as	**elif**	**if**	**or**	**yield**
assert	**else**	**import**	**pass**	
break	**except**	**in**	**raise**	



Identifiers (1)

- In programming languages, identifiers are names used to identify a variable, function, or other entities in a program.
- The **rules for naming an identifier in Python** are as follows:
    1) The name should begin with an uppercase or a lowercase alphabet or an underscore sign (_). This may be followed by any combination of characters **a–z**, **A–Z**, **0–9** or underscore (_).
        ➔ An identifier CANNOT start with a digit.



Identifiers (2)

2) It can be of any length. (However, it is preferred to keep it short and meaningful).
3) It should not be a keyword or reserved word given in previous table.
4) We cannot use special symbols like !, @, #, $, %, etc., in identifiers.



Variables

- A variable in a program is uniquely identified by a name (identifier).
- Variable in Python refers to an object — an item or element that is stored in the memory.
- Value of a variable can be a string (e.g., 'b', 'Global Citizen'), numeric (e.g., 345) or any combination of alphanumeric characters (CD67).
- In Python we can use an assignment statement to create new variables and assign specific values to them.



Variables in Python

- Variable declaration is implicit in Python, means variables are automatically declared and defined when they are assigned a value the first time.
- Variables must always be assigned values before they are used in expressions as otherwise it will lead to an error in the program.
- Wherever a variable name occurs in an expression, the **interpreter** replaces it with the value of that particular variable.



Variables: Try out (1)

- Write a program to display "keep smiling" assigned to a variable in Python.
- Write a program to display "The user number is 101", with 101 is assigned to a variable in Python.

```
>>> message="Keep Smiling"
>>> print(message)
Keep Smiling
>>>
```

```
>>> userNo=101
>>> print('The user number is',userNo)
The user number is 101
>>>
```



Variables: Try out (2)

- Write a Python program to find the area of a rectangle given that its length is 10 units and breadth is 20 units.

```
>>> length = 10
>>> breadth = 20
>>> area = length * breadth
>>> print(area)
200
>>> print('The area of the rectangle is',area)
The area of the rectangle is 200
>>>
```

Comments in Python

- Comments are used to add a remark or a note in the source code.
- Comments are not executed by interpreter.
- In Python, a comment starts with # (hash sign).
- Everything following the # till the end of that line is treated as a comment and the interpreter simply ignores it while executing the statement.

```
>>> #To find the sum of two numbers
>>> num1 = 10
>>> num2 = 20
>>> result = num1 + num2
>>> print(result)
30
>>>
```

Everything is an Object

- Python treats every value or data item whether numeric, string, or other type (discussed in the next topic) as an object in the sense that it can be assigned to some variable or can be passed to a function as an argument.
- Every object in Python is assigned a unique identity (ID) which remains the same for the lifetime of that object. This ID is akin to the memory address of the object.

  ➔ The function id() returns the identity of an object.

```
>>> num1 = 20
>>> #Identity of num1
>>> id(num1)
1651325102992
>>> num2 = 30 - 10
>>> #Identity of num2 and num1
>>> #are same
>>> #as both refers to object 20
>>> id(num2)
1651325102992
>>>
```



1) List any four Python IDEs.
2) What are the basic modes of python?
3) What is Python Shell or Python Interactive Shell?
4) What is script mode?
5) Differentiate between script mode and interactive mode.
6) What is Jupyter Notebook?
7) What is the extension of Python program/script?



Any Questions?

Thank you for listening ☺

